

Andrew Chen

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EDUCATION

University of California, Davis

Expected Graduation: June 2027

B.S. in Statistics & B.A. in Economics

- GPA: 3.94/4.00
- Dean's Honor List: Fall 2023, Winter 2024, Spring 2024, Spring 2025, Winter 2026

PROFESSIONAL EXPERIENCE

AI Research & Product Development, Intelligent Asset Management Intern — JD.com

Jul 2026 – Present

- Built and deployed an AI-powered monitoring skill combining statistical anomaly detection with grounded LLM reporting, achieving 98% recall and 80% precision while replacing weekly manual reviews with automated T+1 alerts and surfacing operational issues up to 7 days earlier.
- Developed a RAG pipeline integrating domain knowledge sources, embedding-based retrieval, and context injection to ground financial-agent responses in internal knowledge, reducing hallucination rate by 3 percentage points and improving response reliability.
- Designed and analyzed an A/B experiment evaluating a hardship-aware repayment strategy across 500K+ delinquent accounts, measuring 30-day cure rate with complaint and policy-error rates as guardrails; identified a 14.6% lift in 30-day cure rate, informing rollout to eligible customers.

AI Research Fellow — UC Davis Department of Economics

Mar 2026 – Jul 2026

- Designed and implemented a GPT-based information extraction pipeline to parse 450+ FOMC meeting transcripts, producing 54,930 structured causal relationships between economic entities as directed graph inputs for a macroeconomics research paper.
- Engineered prompt frameworks with predefined schemas to constrain LLM output to 35 economically relevant entity types, enforcing structured generation and reducing out-of-schema noise to under 1% of extracted relationships.
- Built data pipelines supporting a Retrieval-Augmented Generation (RAG) system, integrating LLMs for downstream analysis; applied NLP techniques and semantic matching to normalize entities into controlled vocabularies to support downstream graph construction and clustering.

Research Assistant — UC Davis Department of Economics

Sep 2025 – Jun 2026

- Built an end-to-end pipeline (Selenium, Tesseract OCR, Gemini API) extracting 100k+ historical central bank gold reserve records from scanned PDFs into analysis-ready datasets, with an optimization algorithm cutting required file retrievals by 65% while preserving 100% data completeness.
- Implemented automated validation with cross-checks and anomaly detection; engineered Gemini prompt templates for structured outputs, improving extraction consistency.

Student Analyst — UC Davis Technology Transfer Office

Aug 2025 – Present

- Managed invention disclosure intake and compliance verification, maintaining a structured patent database.

Business Analyst Intern — PAGO LLC

Jul 2025 – Sep 2025

- Queried and analyzed operational datasets (5,000+ SKUs) via SQL/Excel to surface workflow bottlenecks; recommendations adopted by management, driving a 20% reduction in order processing time.

PROJECTS

AI Investment Copilot | Python, FastAPI, Gemini API, Redis, Docker, GitHub Actions, Pydantic

- Built a production-style agentic financial analysis system: custom ReAct agent (Thought → Tool Call → Observation → Answer) orchestrating a Pydantic-validated tool registry over real-time market data and quant risk calculations, served via FastAPI with SSE streaming.
- Developed an interactive Streamlit dashboard with technical-analysis visualizations, options payoff charts, live agent activity, and research briefs for cash-secured puts, volatility analysis, and multi-stock comparisons.
- Created a layered evaluation system spanning pytest unit tests, 10 golden-query workflows, deterministic tool/answer contracts, LLM-as-judge scoring, SQLite persistence, and automated Markdown reports—achieving a 100% golden-evaluation pass rate and 5.0/5.0 judge score.
- Engineered Redis-backed caching, rate limiting, and session history; added trace-ID observability, structured logging, bounded timeout retries, Docker Compose deployment, and GitHub Actions CI for automated testing and container validation.

FinSight | Python, FastAPI, OpenAI API, Cohere Rerank, BM25, Docker, GitHub Actions, Pydantic

- Built an end-to-end RAG system over SEC 10-K filings, ingesting raw EDGAR HTML/XBRL and applying a custom Item- and sentence-aware chunker that preserves section provenance and produces citation-ready excerpts.
- Implemented four retrieval modes—dense, BM25, RRF hybrid, and hybrid with Cohere reranking—improving Fact Recall@5 from 0.71 to 0.87 and Full Coverage@5 from 0.40 to 0.70 versus dense retrieval while achieving required-evidence hits for all 10 benchmark questions.
- Designed a source-grounded evaluation framework with 45 required facts, 55 optional facts and 110 human-verified quotes; built a deterministic resolver that automatically generates chunk mappings, character offsets, and SHA-256 fingerprints instead of manually selected chunk IDs.
- Built a GPT-4o-mini answer-generation layer with citations traced to ticker, filing section, and source URL; exposed it through FastAPI and a Streamlit debugging UI, containerized both services with Docker Compose, and configured GitHub Actions to run 50 pytest tests on every push.

SKILLS

- Languages:** Python, R, SQL
- ML / Deep Learning:** Supervised & unsupervised learning; model optimization, regularization, and evaluation (ROC-AUC, F1, MAE); scikit-learn; PyTorch (CNNs, LSTMs, Transformers); dimensionality reduction (PCA), clustering
- AI / LLM Systems:** Agentic systems (ReAct); RAG (hybrid dense + BM25 retrieval, reciprocal rank fusion); LLM-as-judge evaluation; prompt design & structured output enforcement; schema validation (Pydantic v2); OpenAI & Gemini API integration
- Infrastructure:** FastAPI, PostgreSQL, Redis, Docker/Docker Compose, AWS (S3, EC2), GitHub Actions (CI/CD), SQLite, Pinecone, Selenium, Tesseract OCR, Git/GitHub, Streamlit